



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018-0200



Agricultural Machinery – Glass Bottle Crusher DP GBC-01

Test Report Number	2018-0200
Date of Issuance	April 30, 2019
Valid Until	April 30, 2024
Agricultural Machinery	Bottle Crusher
Type	Stationary, rotary, cylindrical
Brand	DP
Model	GBC-01
Test Requested by	DP FABRICATION AND MACHINERIES San Miguel Road San Jose, Digos City, Davao Del Sur

SUMMARY

Performance Criteria	Manufacturer's Specifications*	AMTEC Test
Input capacity, kg/h	ND	617.84
Noise level, db(A), maximum	ND	113
Crushing Efficiency, %, minimum	ND	96.18
Fineness Modulus	ND	4.47

Note: *Manufacturer's specifications were not provided

OBSERVATIONS

1. Prime Mover	3.73 kW MINDONG Electric Motor
2. Operator's Manual	None
3. Previous Valid Test Reports of Similar Brand and Model	None
4. Manufacturer	DP FABRICATION AND MACHINERIES San Miguel Road San Jose, Digos City, Davao Del Sur
5. Belt-and-Pulley Transmission	With cover
6. Number of Operators	2 Feeder, 1 Bagger
7. Loading of Input	Accessible to the operator
8. Adjustments	No adjustments were done during operation.
9. Collection of Output	Manually collected with the use of container
10. Safety Features	Crushing chamber was fully covered during operation; hence, no materials spilled out or scattered that can injure the operators.
11. Other Features	Safety goggles were worn during operation. The machine tends to stop when thick bottles were loaded (e.g. softdrink bottles).

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	April 22, 2018
Location of Test	Brgy. San Jose, San Miguel Road, Digos, Davao del Sur
Items Tested	Machine specifications, Input capacity, Speed of components, Noise level, Power consumption, and Labor requirement

METHODS OF TEST

The glass bottle crusher was tested based on the “AMTEC– Methods of Test”.

DESCRIPTION OF THE MACHINE



Figure 1. DP GBC-01 Glass bottle crusher

- a. Hopper
- b. Crushing Chamber
- c. Discharge Output

SPECIFICATIONS

Item	Manufacturer's Specification	AMTEC Verification
1 Main structure		
1.1 Overall dimensions, mm		
1.1.1 Length	ND	1420
1.1.2 Width	ND	910
1.1.3 Height	ND	1610
1.2 Weight, without the prime mover, kg, if applicable	ND	NM
2 Loading hopper		
2.1 Material	ND	1.0 mm G.I. Sheet
2.2 Opening, mm		
2.2.1 Length	ND	290
2.2.2 Width	ND	260
2.3 Height from the ground, mm	ND	1480
3 Crushing assembly		
3.1 Crusher cover material	ND	4 mm B.I. Sheet
3.2 Cylinder		
3.2.1 Material	ND	14 mm round bar, square bar
3.2.2 Dimensions, mm		
3.2.2.1 Length	ND	ND
3.2.2.2 Diameter	ND	520
3.3 Rotating Shaft	ND	NM
3.3.1 Material	ND	31.75 mm CRS
3.3.2 Dimensions, mm		
3.3.2.1 Length	ND	ND
3.3.2.2 Width	ND	3.175
3.3.3 No. of Crushers	ND	16
3.3.3.1 Counter Crushers	ND	15
3.3.3.2 Material	ND	Plain round bar
3.3.3.3 Dimensions, mm		
3.3.3.4 Length	ND	ND

Note: ND-No Data, NM-Not Measured

SPECIFICATIONS (Cont'n)

Item	Manufacturer's Specification	AMTEC Verification
3.3.3.5 Diameter	ND	14 mm
4 Output chute		
4.1 Material	ND	1.0 mm GI Sheet
4.2 Dimensions, mm	ND	
4.2.1 Length	ND	130
4.2.2 Width	ND	100
4.3 Height from the ground	ND	570
5 Safety features	ND	ND
6 Chassis		
6.1 Material	ND	50 mm x 50 mm x 3 mm Angle bar
7 Perforated Bed		
7.1 Dimensions, mm		
7.2 Diameter of Perforations	ND	12
7.2.1 Length, mm	ND	ND
7.2.2 Width, mm	ND	76.2
7.2.3 Thickness, mm	ND	6.35
8 Prime Mover (Electric Motor)		
8.1 Brand	ND	MINDONG
8.2 Model	ND	YC132M-4
8.3 Make	ND	China
8.4 Serial number	ND	ND
8.5 Type	ND	Single-phase Induction
8.6 Rated power, kW	ND	3.73
8.7 Rated speed, rpm	ND	1740
8.8 Cooling system	ND	Air-cooled
8.9 Power transmission system	ND	Belt and Pulley
8.9.1 Engine pulley diameter, mm	ND	10.16
8.9.2 Crusher pulley diameter, mm	ND	30.48
8.9.3 No. of pieces of belt	ND	1

Note: ND-No Data, NM-Not Measured

RESULTS**A. PERFORMANCE TEST**

Item	Data	
1 Information of test materials		
1.1 Material	Assorted glass bottles	
1.2 Source	ND	
1.3 Dimensions, mm		
1.3.1 Length	NM	
1.3.3 Thickness	NM	
2 Weight of input, kg	51.9	
3 Operating time, h	0.084	
4 Input capacity, kg/h	617.86	
5 Crushing efficiency, %	96.18	
6 Speed of components, rpm	With Load	Without Load
6.1 Electric motor shaft	1788	1805
6.2 Crusher shaft	574	606
6.3 Feeder	109.4	74.9
6.4 Bagger	113	81.8
7 Prime Mover		
7.1 Voltage	229.7	225
7.2 Current	22.2	7
7.3 Power (kW)	3.0	1.6
8 Output Fineness Modulus for Crushed Bottle	4.47	

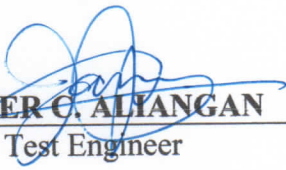
B. LABORATORY ANALYSIS (FINENESS MODULUS)

Sieve Mesh Number	Wt. of Material Retained, g	Percent of Material Retained, %	Multiplier	N
12	0	0.00	7	0.00
16	43.70	43.64	6	261.84
20	16.94	16.92	5	84.58
30	15.37	15.35	4	61.41
40	7.51	7.50	3	22.50
50	5.74	5.73	2	11.46
100	5.62	5.62	1	5.62
Pan	5.25	5.24	0	0.00
PTotal	100.13	100.00		447.41
Fineness Modulus	4.47			
Particle size, Diameter, mm	1.091			

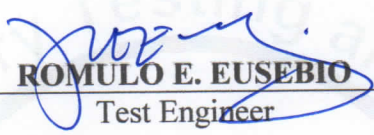
OTHER OBSERVATIONS

Figure 2. Pulverized Glass by DP GBC-01

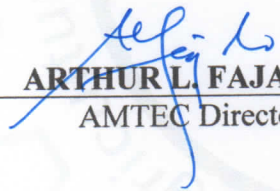
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MAY 06 2019
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